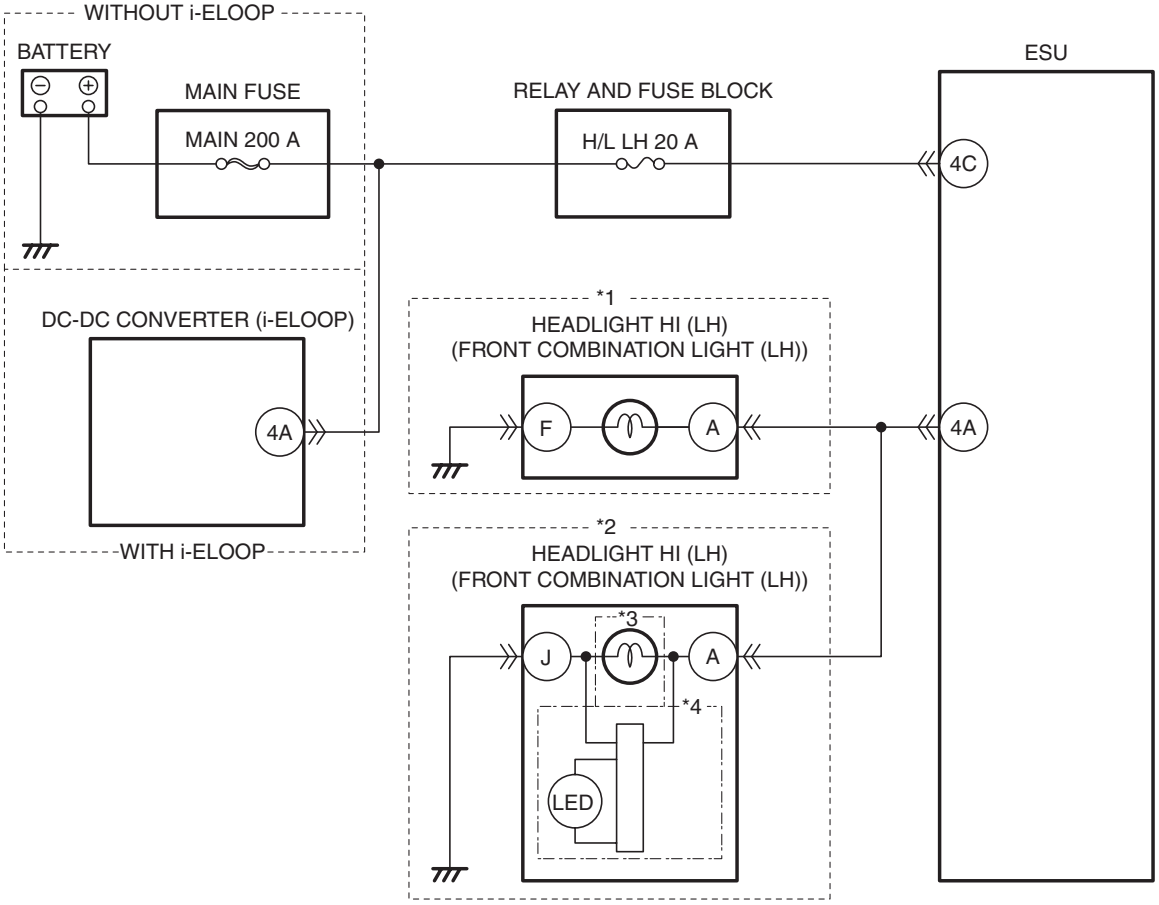
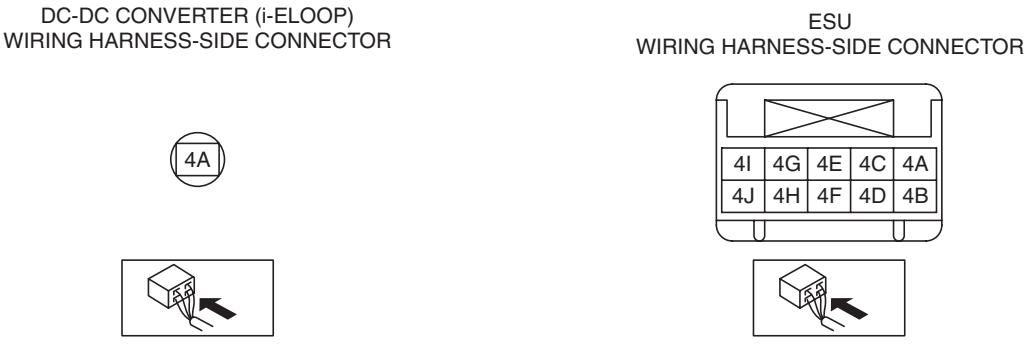
DTC B1D02:11 [ELECTRICAL SUPPLY UNIT (ESU)]

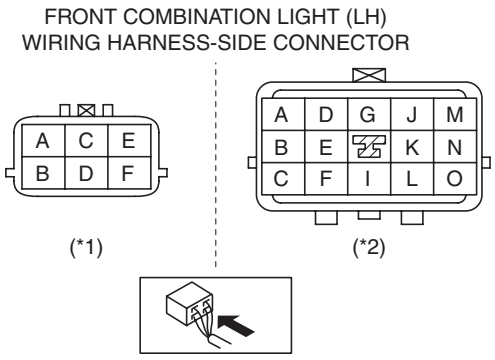
id0902ff062500

Description	Headlight HI (LH) circuit malfunction
Detection condition	• Electrical supply unit (ESU) detects a short to ground in the headlight HI (LH) circuit.
Fail-safe	Not applicable
Possible cause	• Electrical supply unit (ESU) connector or terminal malfunction • Headlight HI/LO bulb (LH) malfunction • Front combination light (LH) connector or terminal malfunction • Short to ground in wiring harness between front combination light (LH) terminal A and electrical supply unit (ESU) terminal 4A • Front combination light (LH) malfunction • Electrical supply unit (ESU) malfunction

Description	Headlight HI (LH) circuit malfunction







*1 : With front combination light (bulb type) and without headlight manual leveling system

*2 : With front combination light (LED type) or headlight manual leveling system

*3 : With front combination light (bulb type)

*4 : With front combination light (LED type)

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the electrical supply unit (ESU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	**With front combination light (bulb type):** • Go to the next step. **With front combination light (LED type):** • Go to Step 3.
		No	Repair or replace the connector, then go to Step 6.
2	**VISUALLY INSPECT HEADLIGHT HI/LO BULB (LH)** • Visually inspect the headlight HI/LO bulb (LH). • Is the headlight HI/LO bulb (LH) normal?	Yes	Go to the next step.
		No	Replace the headlight HI/LO bulb (LH), then go to Step 6. (See HEADLIGHT BULB REMOVAL/INSTALLATION.)
3	**INSPECT FRONT COMBINATION LIGHT (LH) CONNECTOR CONDITION** • Disconnect the front combination light (LH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
4	**INSPECT HEADLIGHT HI (LH) CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the front combination light (LH) and electrical supply unit (ESU) connectors are disconnected. • Inspect for continuity between front combination light (LH) terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between front combination light (LH) terminal A and electrical supply unit (ESU) terminal 4A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 6.
		No	**With front combination light (bulb type):** • Go to Step 6. **With front combination light (LED type):** • Go to the next step.

Step	Inspection		Action
5	**VERIFY IF MALFUNCTIONING LOCATION IS FRONT COMBINATION LIGHT (LH) DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Operate the light switch to the HEAD position and the dimmer switch to HI position. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Replace the front combination light (LH), then go to the next step. (See FRONT COMBINATION LIGHT REMOVAL/ INSTALLATION.)
		No	Go to Step 7.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Operate the light switch to the HEAD position and the dimmer switch to HI position. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].)
		No	DTC troubleshooting completed.